A
MINI
PROJECT
on

Line Following Car

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By

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ABSTRACT

Line Following is one of the most important aspects of robotics. A Line Following Robot is an autonomous robot which is able to follow either a black line that is drawn on the surface consisting of a contrasting color. It is designed to move automatically and follow the line. The robot uses arrays of optical sensors to identify the line, thus assisting the robot to stay on the track. The array of four sensor makes its movement precise and flexible. The robot is driven by DC gear motors to control the movement of the wheels. This project aims to implement the algorithm and control the movement of the robot by proper tuning of the control parameters and thus achieve better performance. It can be used industrial automated equipment carriers, small household applications, tour guides in museums and other similar applications, etc.

ACKNOWLEDGEMENT

I would like to heartedly express my thanks to Mr. Prateek Gupta, Project Supervisor, Department of Mechanical Engineering, for sparing his valuable time to extend help in every step of our project work and give his valuable suggestions to make this project and report a success. I mainly indebted to the authors of many references and articles which were used as the reference. Last but not the least i would like to thank my friends and family for their help in every way for the success of this project report.

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SIGNATURE

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INTRODUCTION

Line follower is a machine that can follow a path. The path can be visible like a black line on a white surface. Sensing a line and maneuvering the robot to stay on course, while constantly correcting wrong moves using feedback from the sensor forms a simple yet effective system. It can be used in automobile, industrial automations, guidance, etc.

As technology becomes increasingly important in today's world, it is invaluable to not only learn how to use technology, but also to understand how to create it. Since being the engineer one should have sound knowledge of the other discipline. Most of the projects have limited scope to only specific discipline. This would limit ones innovation and creativity. This project inspires to make connections across several disciplines rather than learning topics in isolation as it combines mechanical, electronic, electrical and programming skills.

- It gives visual grasp of math and science.
- It builds logical thinking.
- It brings out innovation and creativity.
- It enhances problem solving skills.

Problem Description

In the industry carriers are required to carry products from one manufacturing plant to another which are usually in different buildings or separate blocks. Conventionally, carts or trucks were used with human drivers. Unreliability and inefficiency in this part of the assembly line formed the weakest link. The project is to automate this sector, using carts to follow a line instead of laying railway tracks which are both costly and an inconvenience.

Objectives

The objectives of the project are:

- The robot must be capable of following a line.
- It should be capable of taking various degrees of turns.
- The robot must be insensitive to environmental factors such as lighting and noise.
- It must allow calibration of the line's darkness threshold.
- Scalability must be a primary concern in the design.

Methodology

- After the detail literature survey through the books, periodical, journal, magazine, websites. The idea of the project is well defined.
- The logic is derived for the intelligence of the robot.
- The accuracy and viability of the electronic components is tested in the simulation software Proteus.
- After the successful simulation result it is implemented in the hardware.
- After the finishing the electrical and electronics part, the stable,reliable and flexible mechanical design and fabrication is completed.
- Finally system is tested and encountered error is omitted.

Limitations

The system has restricted to the following limitation.

- Choice of line is made in the hardware abstraction and cannot be changed by software.
- Few curves are not made efficiently, and must be avoided.
- The turning radius should be of minimum 100m to take smooth U-turning of robot.

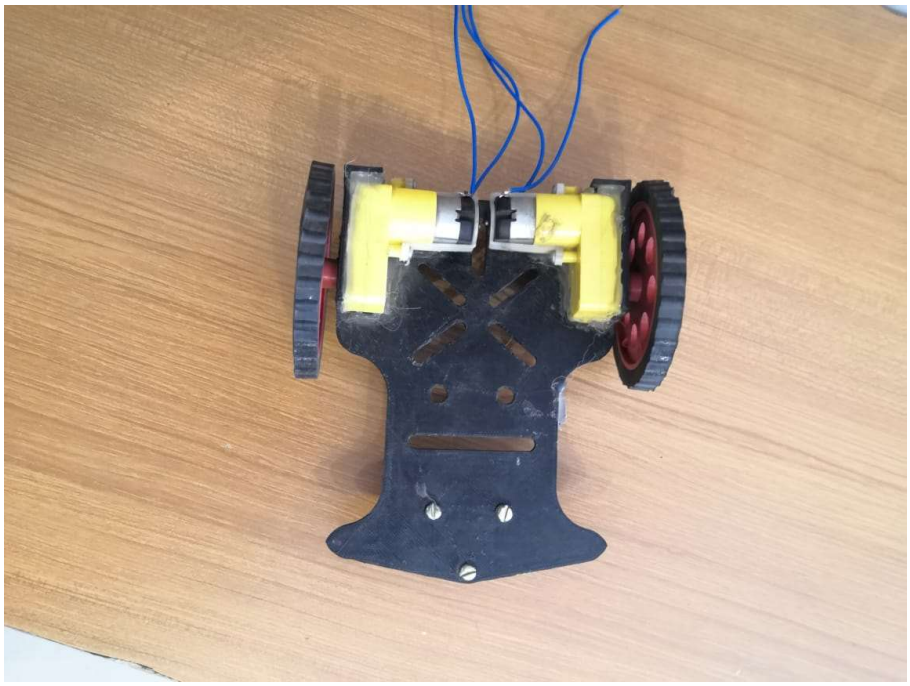
- The width of the path must be of 45mm so that it can cover minimum 3 sensors.
- The path should be plane and obstacle free.
- The steering mechanism is not easily implemented in huge vehicles and impossible for non-electric vehicles.

Hardwares Required

The hardware required is divided in the following category:

Basic Components

- **Chasis**



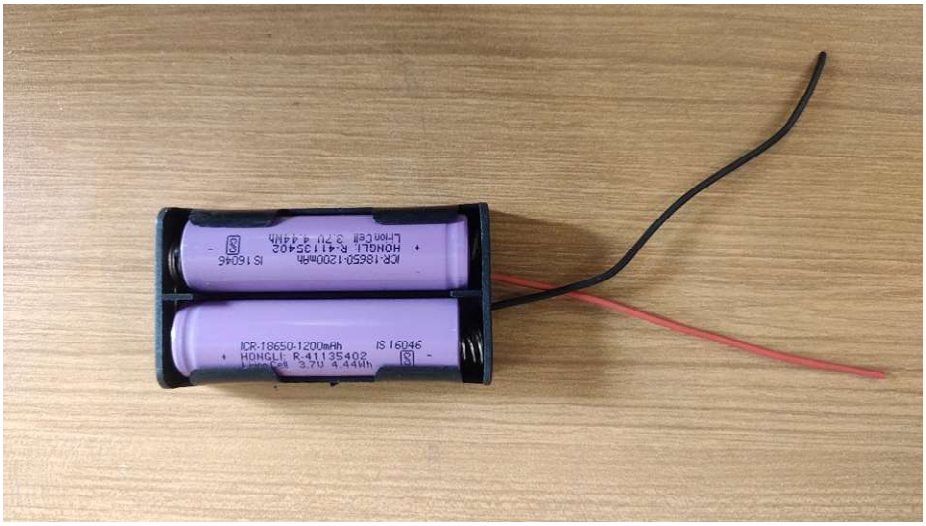
- **Wheels**

  Here, we use 2 wheels of following kind.

- **Caste**

Power Components

- **7.4 V Batteries**
  Two Batteries of 7.4 V are used in this Line Follower each having a potential difference of 3.7 V.



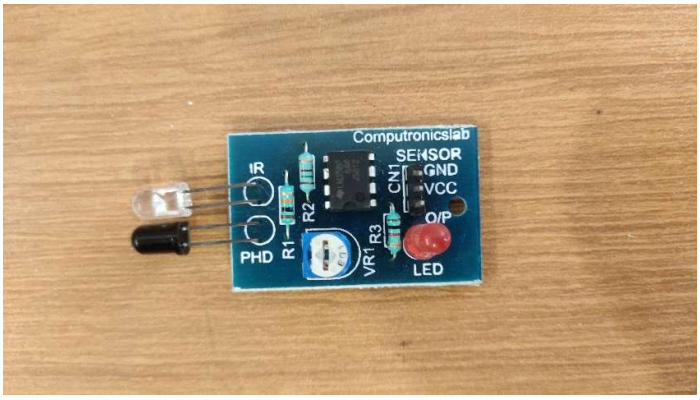
- **ON/OFF Switch**

Input System

The detail of each components is discussed below.

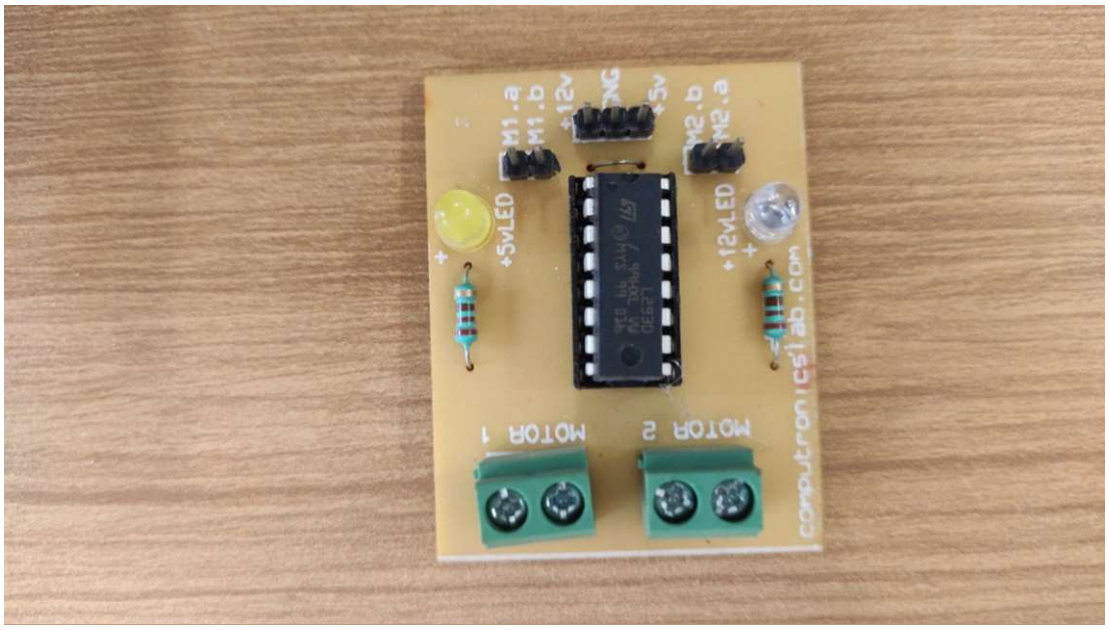
- Proximity Sensor

  The combination of IR- LED and Photodiode is used as the reflective optical sensor. It generate interrupt when the IR-beam is break to the photodiode. To create the IR breakbeam, IR LED is used with a low value resistor so that it shines very bright. The receiver is Photodiode which biases 'on' whenever the IR LED's light is detected. A sensor will be placed adjacent the IR link and turned on so as to generate a pulse to the Motor Driver.

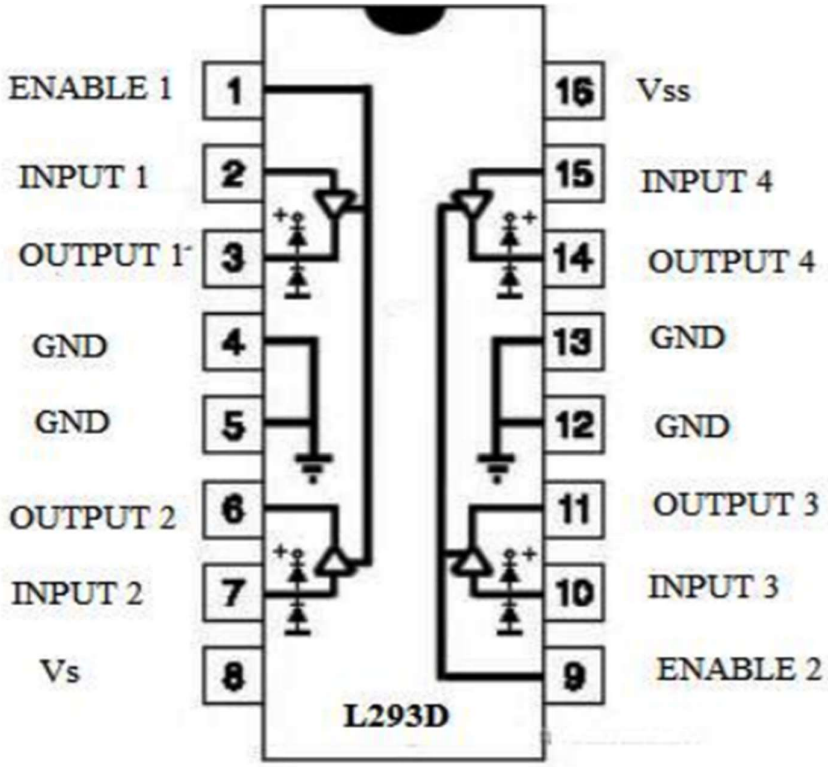


- **Motor Driver**

  Motor driver is a current enhancing device; it can also be act as Switching Device. Thus, after inserting motor driver among the motor and microcontroller. Motor driver taking the input signals from microcontroller and generate corresponding output for motor. IC L293D This is a motor driver IC that can drive two motor simultaneously. Supply voltage (Vs) is the voltage at which motor drive. Generally, 6V for dc motor and 6 to 12V for gear motor are used, depending upon the rating of the motor. Logical Supply Voltage deciding what value of input voltage should be considered as high or low .So if the logical supply voltage equals to +5V, then -0.3V to 1.5V will be considered as Input low voltage and 2.3V to 5V is taken into consider as Input High Voltage. The Enable 1 and Enable 2 are the input pin for the PWM led speed control for the motor L293D has 2 Channels .One channel is used for one motor.



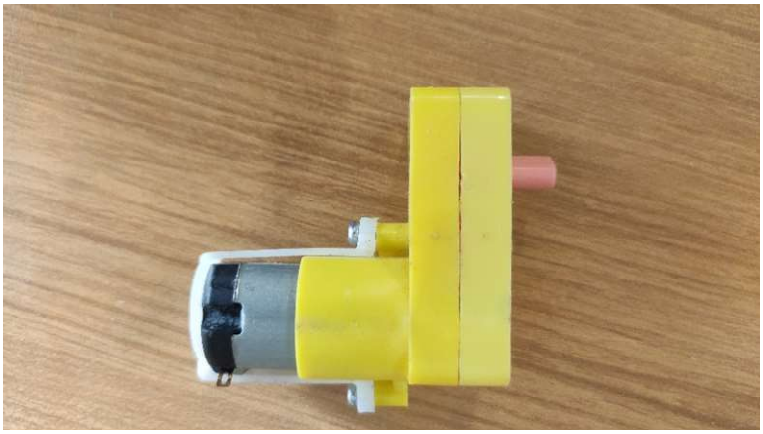
Pin Con



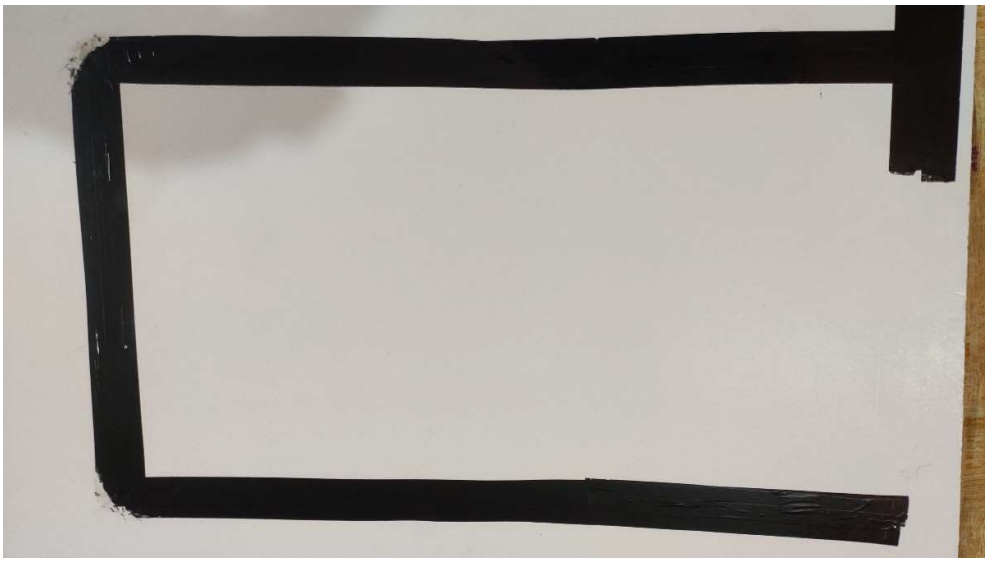
- **DC Motor (6-9 V)**

  Motor is a device that converts any form of energy into mechanical energy or imparts motion. In constructing a robot, motor usually plays an important role by giving movement to the robot. In general, motor operating with the effect of conductor with

current and the permanent magnetic field. The conductor with current usually producing magnetic field that will react with the magnetic field produces by the permanent magnet to make the motor rotate. There are generally three basic types of motor, DC motor, even servomotor and stepper motor, which are always being used in building a robot. DC motors are most easy for controlling. One DC motor has two signals for its operation. Reversing the polarity of the power supply across it can change the direction required. Speed can be varied by varying the voltage across motor.



- **Track**



ARRANGEMENT

- **Sensors' arrangement**

For successful navigation on critical paths, number of sensor and their position play vital role. In general, a matrix of sensors with eight sensor or at least six sensor make a robot able to detect critical junctions. In our robot we set different experiments with sensors to reduce its number. To detect any line minimum two sensors are required. Using these two sensor a robot can easily follow a straight line or simple wavy line. In our robot we used an array of only four sensor, where middle two sensors were set for follow simple line. We set this two sensor such that in ideal position the line remain just between these sensors. But using only two sensor it is not possible to detect a T-junction, + junction or any 90 angle. So, we add additional two sensor on both sides, which helped our robot to detect T-junction and 90 angle. Though, using only four sensor our robot still not able to directly detect a + junction. But, we developed a feedback based algorithm such that, whenever any + junction appears, our robot able to choose the correct path from three alternative options: go straight, take a 90 degree left turn or take a 90 degree right turn.

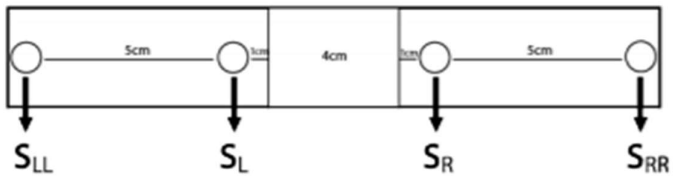
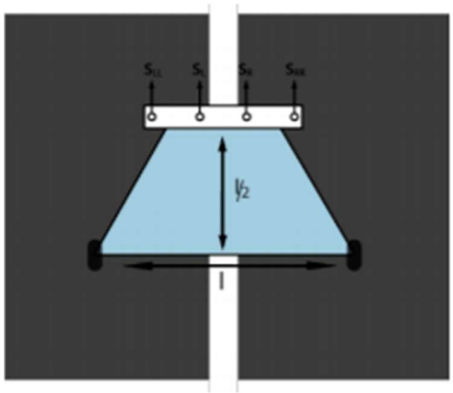
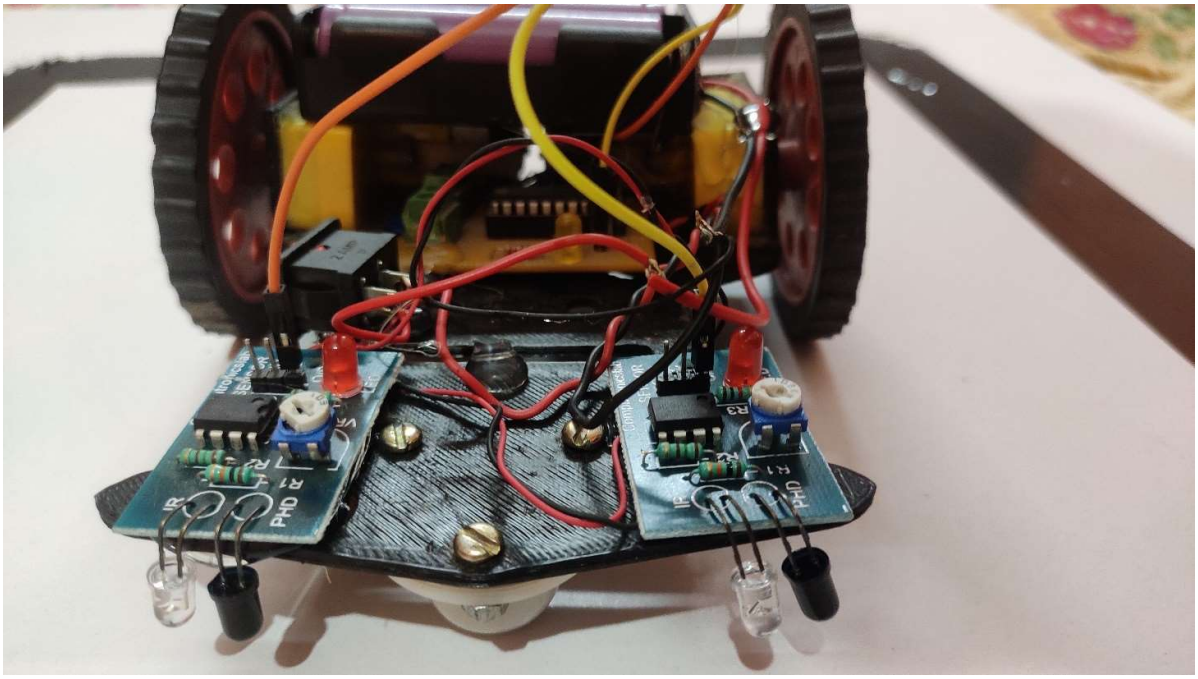


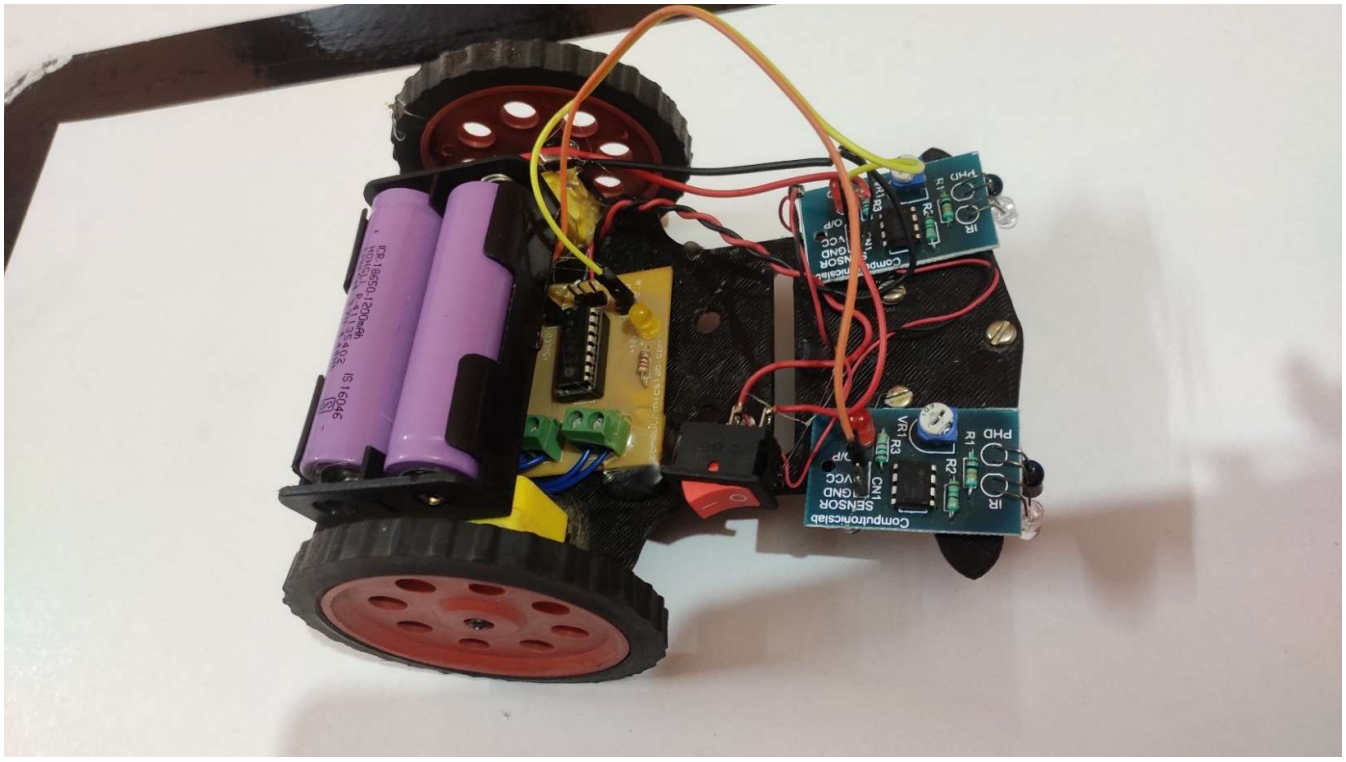
Figure 5: Sensor array

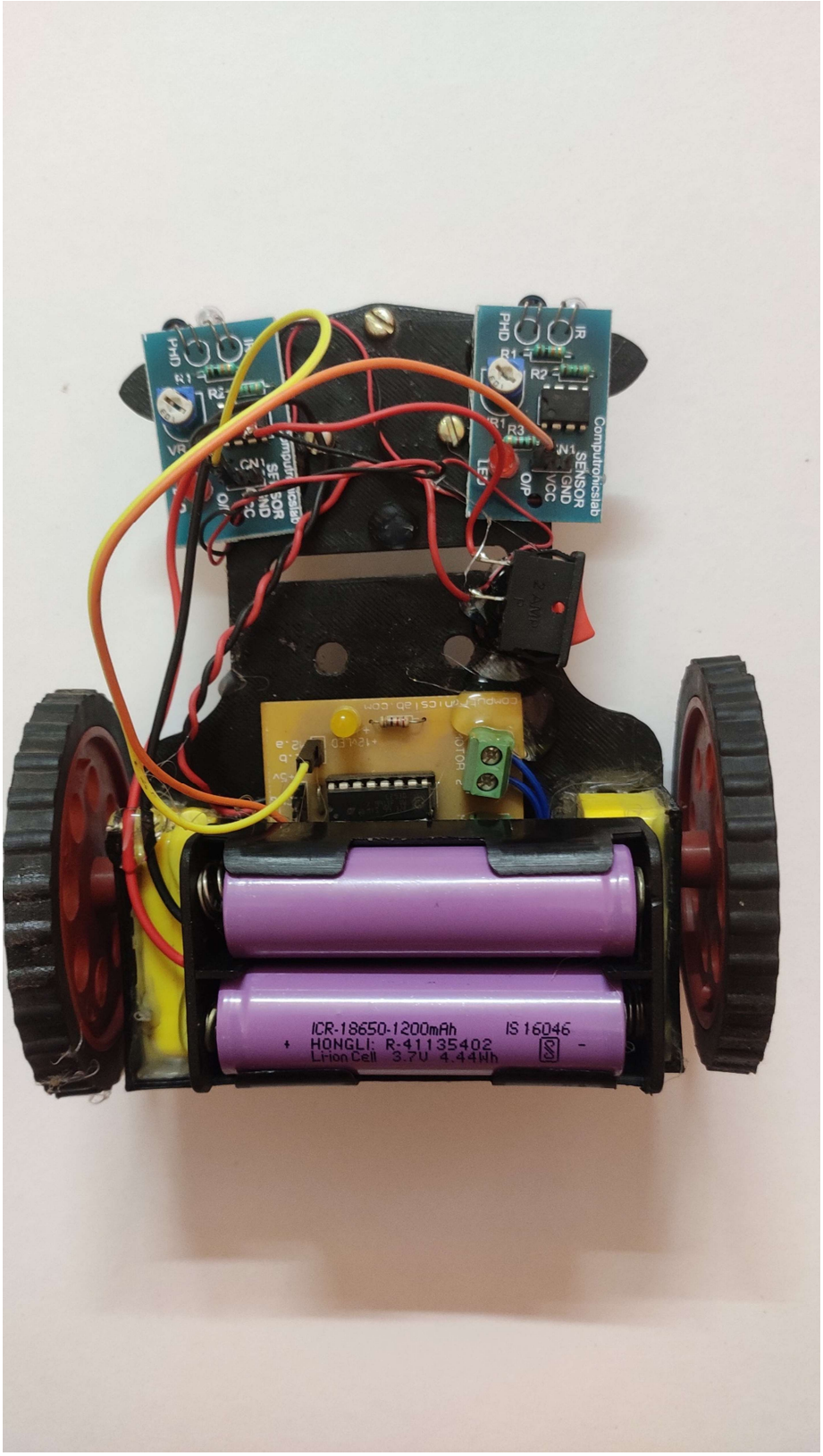


Robot's Block Diagram

Assembly






PHD
IR
R1
R2
R3
VR
Computronicslab
SENSOR
GND
VCC
O/P
ICR-18650-1200mAh
IS 16046
HONGLI: R-41135402
Li-ion Cell 3.7V 4.44Wh

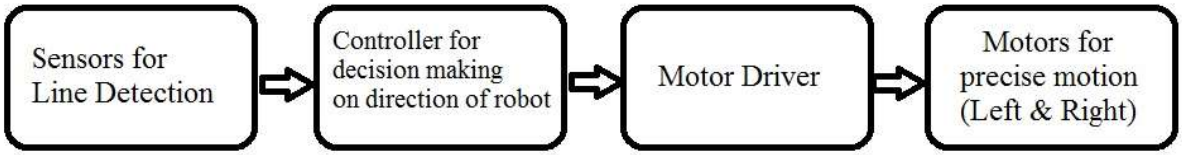
Operation of Line Following Car

The basic operations of line follower are as follows:

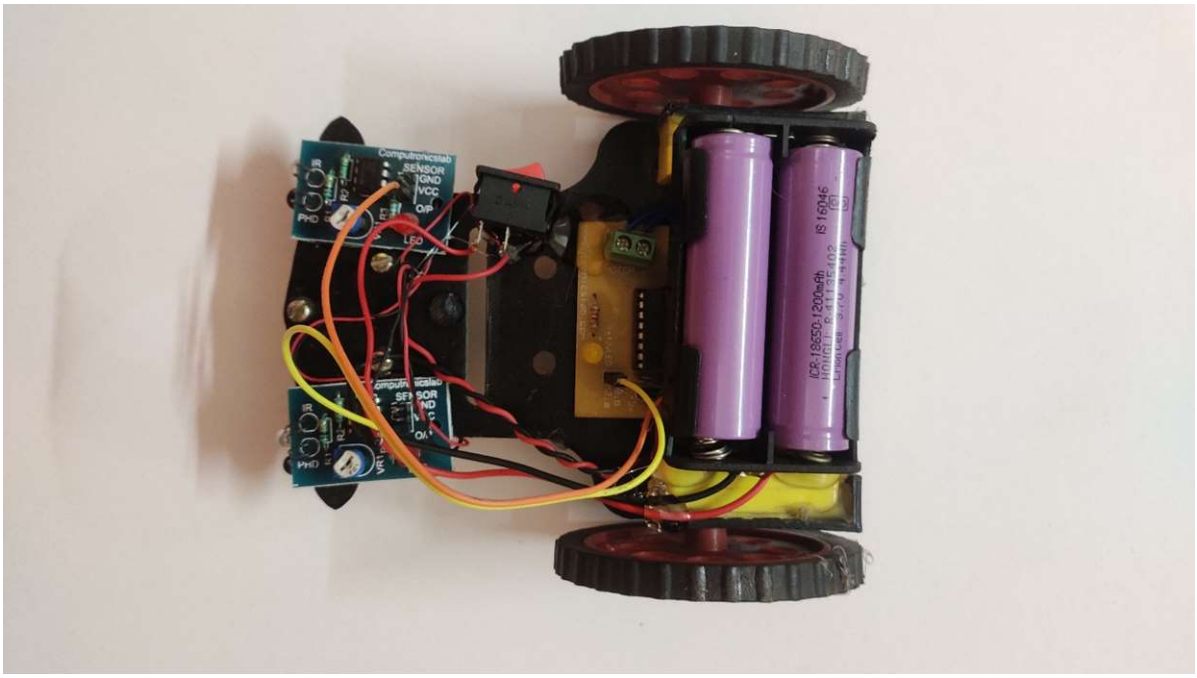
- Capture line position with optical sensors mounted at front end of the robot. For this a combination of IR-LED and Photodiode called an optical sensor has been used. This make sensing process of high resolution and high robustness.

- Steer robot requires steering mechanism for tracking. Two motors governing wheel motion are used for achieving this task.

- On the detecting no black surface robot move in a circular motion until line is found.



Block Diagram for Line Follower Robot



Working Principle

Here middle two sensor SR and SL are used to detect simple line. Outer left sensor SLL were set to detect 90° left turn and outer right sensor SRR to detect 90° right turn. Distance between SR and SL sensors is 6 cm. So, in ideal position these two remained 1 cm far from main base line. Distance between sensors SRR to SR and SLL to SL were set 5 cm for both cases.

- **Line following algorithm**

Following flow-chart describes the basic working principle of our robot.

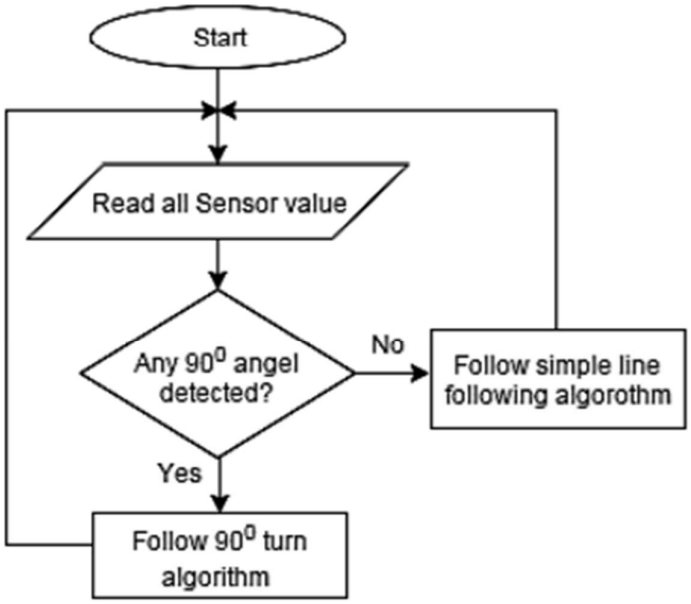
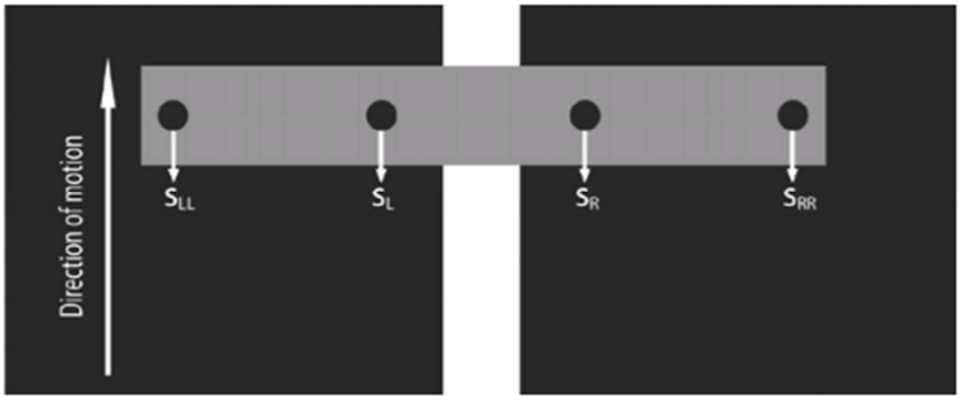


Figure 7: Basic working principle.

As shown on this flow chart, the navigational strategy is divided into two parts: simple line detection and simple line following algorithm, 90° angle detection and 90° turn algorithm.

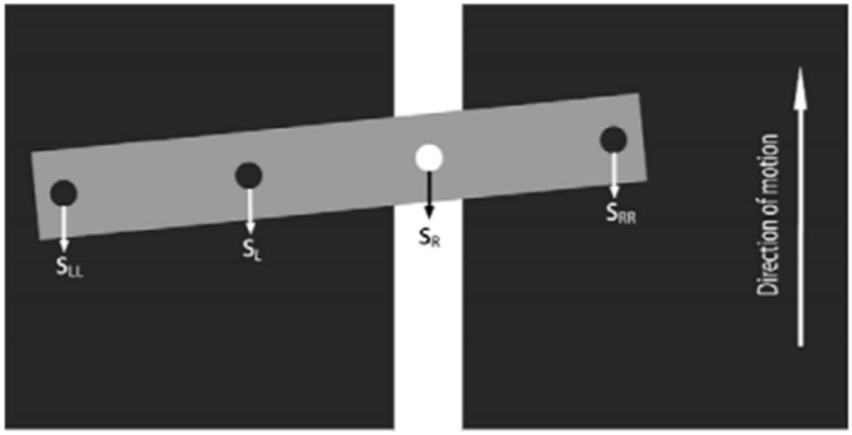
- **Simple line detection and simple line following algorithm**

This robot detect a line as simple line and follow simple line following algorithm if and only if both of its outer sensors (SLL, SRR) are on black surface. Here, three situations can be happened, as shown in figures below.

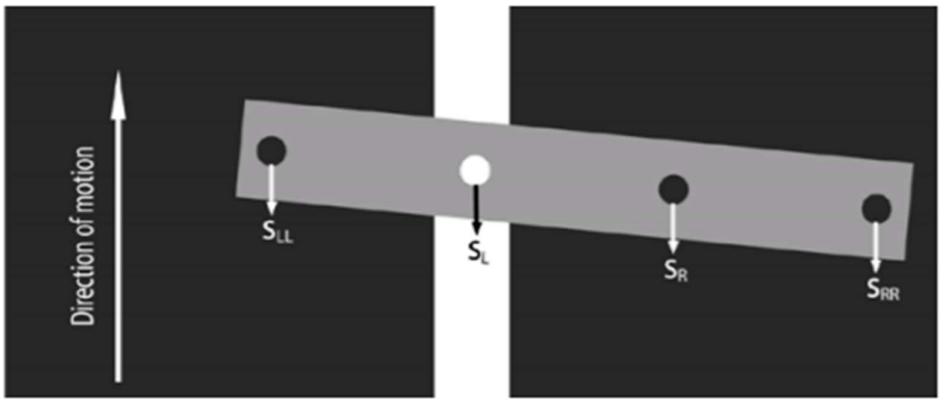


Consider this figure, when all sensors are on bla ‘

ck surface. This the ideal position for the robot. In this situation robot will move forward with full speed by rotating it’s both motors at maximum rpm.



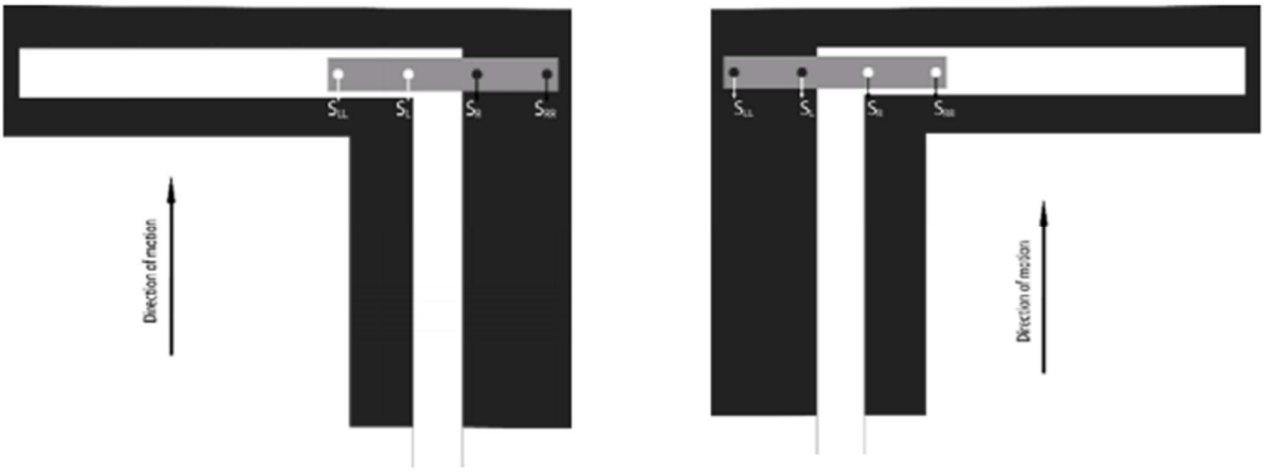
In second situation both outer two sensors (SLL, SRR) and middle left sensor (SL) are on black surface, only the middle right sensor (SR) is on white line. So, the robot need to move right until middle right sensor (SR) backs to black surface (to move back to ideal position). To do so, this robot slow down the right motor speed to half and keep left motor to highest speed. Rather than stopping the right motor, we reduce its speed to gain better navigation.



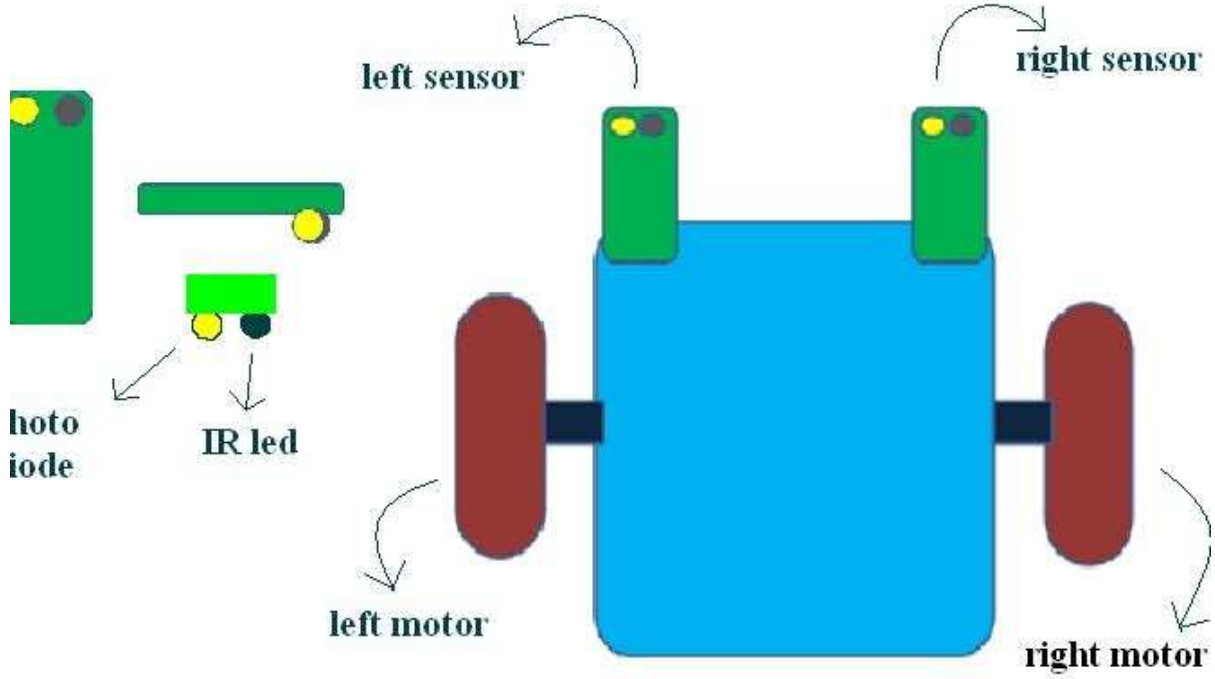
In third situation, middle left sensor (SL) is on white line and other three sensors are on black surface. So, the robot need to move left until middle left sensor (SL) backs to black surface. So, we reduce left motor's speed to half and keep other's high.

- **90 degree angle detection and 90 degree turn algorithm**

  If the robot detects any 90° angle in its path, then it will follow 90° turn algorithm. This can be happen in two ways- 90° left turn or 90° right turn.



As shown in figures , our robot will detect a path segment as 90 degree angle if either both SL and SLL sensors are on white line or both SR and SRR sensors are on white line. If both SL and SLL are on white line, this robot will take a 90 degree left turn. To do so, we rotate left motor in reverse way with full speed and rotate right motor in forward way with full speed, until both SRR and SLL sensor are back on black surface. To take a 90 degree right turn, we do the same thing in reverse way.


left sensor
right sensor
hoto
iode
IR led
left motor
right motor

Results and Discussion

Our robot able to follow all parts of sample test path smoothly. Using above two algorithms, this robot easily able to detect and follow straight line, simple curve and 90° left or right bends. But whenever any T-junction or + junction appears on the path, our robot need additional instruction to choose correct paths from alternatives. A T-junctions opens two alternatives paths to robot: a 90 degree left turn or a 90° right turn. As mention earlier at a + junction the robot has to choose correct path from three alternatives: go straight, take a 90° left turn or take a 90° right turn. If we closely analysis these alternative options from both T-junction and + junction, we can see that our robot can follow any one of these using above two algorithms. It just need to choose correct option. To do so, we have used a counter variable with initial value 0. As, shown in figure, when all sensors were on white line, then robot detected it as a T-junction. Again, for + junction, this robot will detect it as T-junction because it has only an array of four sensors.

Conclusion and Future Scope

Though, the use of line follower robot is very common today. But, select optimal sensors to achieve high efficiency is still now a great challenge. This paper gives a solution on this matter. It discuss about the algorithm through which a line follower robot able to follow any 90° bends, T or + junctions using minimum number of sensors. It also discuss about the number of sensors a robot need to follow a simple line and calculate the minimum number of sensors a robot should have if it have to follow a path with critical sections like 90° turn, T-junction and + junction using this algorithm. Our robot successfully able to follow the sample path. Which proves the effectiveness of our proposed method. However, this robot had some limitations too. When we try to increase robot's speed, some problem occurs on 90° bend and junctions. It was because of the momentum of the robot it got through higher speed. We, solved this problem by reducing robot's speed at bends and junctions. In future we can extend our work on this point to ensure efficiency at higher speed.